

INDIAN GEOGRAPHY

Comprehensive Exam-Oriented Study Notes

Allahabad High Court RO | UPPCS | SSC | UPSC Prelims | State PCS

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CHAPTER 1: INDIA — BASIC FACTS & LOCATION

1.1 India's Position in the World

India lies in the **Northern** and **Eastern** hemispheres of the globe.

Aspect	Details
Latitudinal Extent (Mainland)	8°4'N to 37°6'N
Longitudinal Extent	68°7'E to 97°25'E
North-South Extent	3,214 km
East-West Extent	2,933 km
Lat/Long Difference	Both ~29° (interesting coincidence!)
Hemisphere	Northern + Eastern
Area	~32.8 lakh sq km (7th largest country)
World Position	2nd most populous; 7th largest by area

Why East-West Extent < North-South Despite Equal Degrees?

Longitude lines converge towards the poles. Therefore, 1° of longitude represents a **shorter distance** at higher latitudes than near the equator. India's eastern states are at higher latitudes — so the actual km distance (2,933 km) is less than the north-south extent (3,214 km), even though the angular difference (~29°) is the same.

EXAM FOCUS

Q: Latitudinal/Longitudinal extent of India?

Q: North-South vs East-West extent — which is more and why?

Q: In which hemispheres does India lie?

Trick: Lat difference = Long difference ≈ 29°; but N-S > E-W in km

1.2 India as a Tropical Nation

Despite extending up to 37°N (which lies in the warm temperate zone), India is classified as a **Tropical Nation**. Two key reasons:

Himalayas as a barrier — The Himalayas block cold Siberian (Tertiary) winds from reaching India, preventing the extreme cold typical of other temperate regions at similar latitudes.

Monsoon Winds — India experiences monsoonal rainfall driven by seasonal wind reversal, which is a characteristic feature of Tropical regions, not Temperate ones.

Non-geographic evidence: India's disease patterns, dietary habits, festivals, and cultural traits align with tropical nations globally.

EXAM FOCUS

Q: Is India a Tropical or Temperate nation? Why?

Answer: Tropical — due to Himalayas blocking cold winds + monsoonal climate

1.3 Standard Meridian & Indian Standard Time (IST)

India's longitudinal extent is 29° (~2 hours). This debate is ongoing on whether India needs **two time zones**.

Aspect	Detail
Standard Meridian	82.5°E (82°30'E)
IST = GMT +	+ 5 hours 30 minutes
Passes through	Mirzapur (UP) — also called 'Clock Tower of India'
Reason for single time zone	Avoid administrative complexity, railway signal conflicts, national unity
Countries with similar size & multiple zones	USA (6 zones), Russia (11 zones), China (uses 1 zone despite size)

★ Frequently Asked in Exams

Standard Meridian = 82.5°E passes through Allahabad (Mirzapur district), UP

IST = UTC + 5:30

Arguments FOR two zones: NE states suffer daylight wastage, energy loss, disrupted circadian rhythm

Arguments AGAINST: Railway safety (collision risk), administrative cost, national unity concerns

1.4 Tropic of Cancer — States it Passes Through

The Tropic of Cancer (23.5°N) passes through **8 Indian states**. Remember with mnemonic: '**Mit Ra Par Gam Chha Jha**' (or: M-T-P-R-G-M-C-Z-J)

Sequence (West to East)	State
1	Gujarat
2	Rajasthan (touches southern tip only)
3	Madhya Pradesh
4	Chhattisgarh
5	Jharkhand
6	West Bengal
7	Tripura
8	Mizoram

🔥 EXAM FOCUS

Q: How many states does the Tropic of Cancer pass through? → 8 states

Q: Which states? → Gujarat, Rajasthan, MP, CG, Jharkhand, WB, Tripura, Mizoram

Mnemonic: Gu Ra Ma Chha Jha Pa Tri Mi

Note: The line enters India from west (Gujarat) to east (Mizoram)

1.5 Important Latitudinal Reference Line across India

A useful north-to-south alignment of key cities:

Shimla (Himachal Pradesh) — Northernmost reference

Delhi (National Capital)

Bhopal (MP Capital)

These cities lie on roughly the same meridian — showing India's vast N-S extent

CHAPTER 2: INDIA'S NEIGHBORS & INTERNATIONAL BOUNDARIES

2.1 Neighboring Countries — Quick Overview

Country	Border States	Special Notes
Pakistan	Gujarat, Rajasthan, Punjab, J&K	Longest land border (3,323 km); Radcliffe Line (1947); Kori Creek dispute; Siachin issue
China	J&K, Himachal, Uttarakhand, Sikkim, Arunachal	LAC (Line of Actual Control); Tibet Autonomous Region; Aksai Chin dispute
Nepal	Uttarakhand, UP, Bihar, WB, Sikkim	Open border (Treaty of Sugauli 1816); Kalapani dispute; Nepal-India Treaty
Bhutan	Assam, Sikkim, Arunachal, WB	Friendly relations; Doklam issue (2017); Manas NP spans border
Bangladesh	WB, Assam, Meghalaya, Mizoram, Tripura	Longest border ~4,156 km; Enclave exchange (100th Amendment)
Myanmar	Arunachal, Nagaland, Manipur, Mizoram	Natural border (Patkai Hills, Naga Hills, Mizo Hills, Lushai Hills)
Afghanistan	PoK (via Wakhan corridor)	India-Afg border through Pakistan-Occupied Kashmir
Sri Lanka	(Maritime boundary)	Palk Strait separates the two; Adam's Bridge (Ram Setu)
Maldives	(Maritime boundary)	8° channel between Maldives and Minicoy Island

★ Frequently Asked in Exams

China shares border with 5 Indian states: J&K, HP, Uttarakhand, Sikkim, Arunachal Pradesh

Bangladesh shares border with 5 states: WB, Assam, Meghalaya, Mizoram, Tripura

Nepal shares border with 5 states: Uttarakhand, UP, Bihar, WB, Sikkim

Myanmar shares border with 4 states: Arunachal, Nagaland, Manipur, Mizoram

Pakistan shares border with 4 states/UTs: Gujarat, Rajasthan, Punjab, J&K

2.2 India-Nepal Border: Kalapani Dispute

Treaty of Sugauli (1816) — signed after Anglo-Nepalese War — demarcated the Kali River as Nepal's western boundary. However, the river changed course over time, leading to a **boundary dispute** over the Kalapani region.

Kalapani is located in **Pithoragarh district, Uttarakhand**

Currently administered by India; Nepal claims it in their political maps

Kailash Mansarovar Yatra route passes through this region

India-Nepal border largely follows the **Shivalik Range** (E to W)

2.3 India-Myanmar Border

This is a **natural border** formed by the **Purvanchal (Eastern) Hills**:

Patkai Bum hills — Arunachal Pradesh

Naga Hills — Nagaland (Highest peak: **Mount Saramati**)

Manipur Hills (Mani Hills) — Manipur → Location of **Loktak Lake** and **Keibul Lamjao NP**

Mizo Hills and Lushai Hills — Mizoram (Blue mountains due to blue-colored rocks → **Phawngpui Blue Mountain NP**)

A river originates from Patkai: **Chindwin River** → tributary of **Irrawaddy River** (Myanmar)

EXAM FOCUS

Mount Saramati = Highest peak of Nagaland (on Naga Hills)

Keibul Lamjao NP → only floating national park in the world → on Loktak Lake → famous for Sangai Deer

Irrawaddy Dolphin = Brackish water dolphin | Gangetic Dolphin = Fresh water dolphin

Phawngpui Blue Mountain NP — Mizoram

2.4 India-Bhutan Border

India and Bhutan share a friendly border along Assam, Sikkim, Arunachal Pradesh, and West Bengal.

Manas National Park (Assam) extends into Bhutan as **Royal Manas National Park**

Doklam — A tri-junction area (India-Bhutan-China) that was a major standoff point in 2017

Bhutan is known for its GNH (Gross National Happiness) index and strong environmental commitment

2.5 India-Bangladesh Border: The Enclave Exchange

The **100th Constitutional Amendment Act (2015)** enabled exchange of enclaves between India and Bangladesh.

Bangladesh has the **longest** land border with India (~4,156 km)

India-Bangladesh border passes through WB, Assam, Meghalaya, Mizoram, Tripura

EXAM FOCUS

100th Amendment — for land boundary agreement with Bangladesh (enclave exchange)

Bangladesh has longest land border with India among all neighbors

CHAPTER 3: PHYSICAL DIVISIONS OF INDIA

India's terrain can be divided into six major physiographic divisions:

Division	Key Features	Area
1. Himalayan Mountains	Young fold mountains; N boundary; arc shape	~5 lakh sq km
2. Northern Plains	Alluvial; most fertile; densely populated	~7 lakh sq km
3. Peninsular Plateau	Oldest landmass; stable; 600m avg elevation	~16 lakh sq km
4. Coastal Plains	West & East coasts; fishing; ports	~1.5 lakh sq km
5. Thar Desert	Arid; NW India; NW states	~1.75 lakh sq km
6. Islands	Andaman & Nicobar + Lakshadweep	~8,000 sq km

3.1 The Himalayas

Formation of the Himalayas

The Himalayas were formed by the **collision of the Indian Plate with the Eurasian Plate**. This is a convergent boundary — but unlike typical ocean-continent convergence, here **two continental plates collided**.

~85 million years ago, the Indian Peninsula broke away from Madagascar and started moving northward

The Indian Plate moved at very high speed, causing it to **lock under** the Eurasian Plate (NOT full subduction)

This caused **doubling of Earth's crust** (why there is no volcanic activity in Himalayas but high earthquake risk)

Multiple collision events = **series of parallel ranges** → not one range, but several

Himalayas are **Fold Mountains** (not volcanic)

★ Frequently Asked in Exams

Himalayas = Fold Mountains formed by Indian Plate + Eurasian Plate collision

No volcanoes in Himalayas — because crust doubled (magma cannot rise)

High earthquake risk due to continuing plate movement

Himalayas are youngest mountain range in India (~55 million years old)

Indian Plate was part of Gondwana Land originally

Divisions of the Himalayas (North to South)

Range	Also Known As	Avg Altitude	Key Features
Greater Himalayas	Himadri	6,000+ m	Highest peaks; Mt Everest, K2, Kanchenjunga; perpetual snow
Lesser Himalayas	Himachal	1,000–4,500 m	Shimla, Mussoorie, Nainital, Darjeeling; hill stations

Range	Also Known As	Avg Altitude	Key Features
Outer Himalayas	Shivalik	900–1,100 m	Youngest; youngest sedimentary; prone to erosion

Divisions of Himalayas (West to East)

Region	Location	Key Points
Punjab Himalaya	Between Indus and Sutlej	Dhauladhar, Pir Panjal ranges
Kumaon Himalaya	Between Sutlej and Kali	Nanda Devi (highest peak in India in open area — 7816m)
Nepal Himalaya	Between Kali and Teesta	Mt Everest (8848m), Kanchenjunga
Assam Himalaya	Between Teesta and Dihang	Namcha Barwa peak; syntaxial bend

Important Himalayan Passes

Pass	State/Location	Connects	Importance
Karakoram Pass	Ladakh (J&K)	India–China (Xinjiang)	Highest motorable pass; Silk Route
Zoji La	J&K	Srinagar to Ladakh	NH-1D; strategic pass; closed in winter
Rohtang Pass	HP (between Kullu & Lahaul-Spiti)	Manali to Keylong	Alternate: Atal Tunnel (2020)
Baralacha La	HP	Lahaul to Ladakh	On Manali-Leh Highway
Shipki La	HP	India–China (Tibet)	Sutlej river enters India here
Nathu La	Sikkim	India–China (Tibet)	Reopened for trade 2006
Bomdi La	Arunachal Pradesh	India–China (Tibet)	Strategic importance (1962 War)
Diphu Pass	Arunachal	India–Myanmar–China trijunction	Tri-junction point

EXAM FOCUS

Highest motorable pass in India: Khardung La (Ladakh) at ~5,359m

Atal Tunnel: Opened 2020; bypasses Rohtang Pass; world's longest road tunnel above 10,000 ft

Nathu La: Important for India-China border trade (Sikkim)

Shipki La: Sutlej enters India through this pass

Zoji La: Key pass connecting Kashmir Valley to Ladakh

3.2 Northern Plains (Indo-Gangetic Plains)

Also called: **Great Plains / Aggradational Plains / Northern Plains**. Formed by alluvial deposits of Himalayan rivers over millions of years.

Average elevation: **~200 m** above sea level (Ambala Ridge: 291m — watershed between Indus & Ganga drainage)

Stretches from Punjab in the west to Assam in the east (~2,400 km long, 240–320 km wide)

Types of Land in the Plains

Type	Description	Location/Features
Bhabar	Coarse gravel/pebble belt at foothills	8–16 km wide; rivers disappear underground; dry zone
Tarai	Wet marshy belt south of Bhabar	Streams reappear; dense vegetation; wildlife; good for farming
Bangar	Old alluvium (older deposits, elevated)	Higher areas; less fertile; kankars (lime nodules) present
Khadar	New alluvium (newer deposits, low-lying)	Near rivers; very fertile; floods annually; best farming

★ Frequently Asked in Exams

Bhabar → Tarai → Bangar → Khadar (North to South sequence near Himalayas)

Khadar = most fertile soil — renewed every flood season

Bangar = older alluvium — contains Kankars (impure lime concretions)

UPSC/UPPCS repeatedly asks: What is Bhabar? What is Tarai? Sequence?

Sub-Divisions of Northern Plains (Doabs)

Doab (Interfluvium)	Between Rivers	Location
Sindh Sagar Doab	Indus & Jhelum	Punjab (now Pakistan mostly)
Chaj Doab	Jhelum & Chenab	Punjab
Rechna Doab	Chenab & Ravi	Punjab
Bari Doab	Ravi & Beas	Punjab
Bist Doab / Jullundur Doab	Beas & Sutlej	Punjab (India side)
Yamuna-Ghaggar Doab	Yamuna & Ghaggar	Haryana Plains

Regional Sub-Plains of Gangetic Plains

Region	Location	Rivers/Features
Rohelkhand Plains	W UP — Ramganga basin	Named after Rohilla tribe
Yamuna-Ganga Doab	Between Yamuna & Ganga	Most important agricultural area
Yamuna-Chambal Doab	Between Yamuna & Chambal	Central UP/Rajasthan
Awadh Plains	Between Gomti & Ghaghra	Central UP — Lucknow region
Mithila Plains	Between Gandak & Kosi	Bihar — language: Maithili; Sita Mata connection
Magadh Plains	South of Son River	Bihar — ancient Magadha empire
Par Delta	Lower Ganga plains	West Bengal — deltaic region
Rar Plains	Adjacent to Chhota Nagpur	Laterite deposits; separates plateau from plains
Deltaic Plains	Ganga-Brahmaputra delta	Largest delta in world; Sundarbans

🔥 EXAM FOCUS

Q: What separates Chhota Nagpur Plateau from plains? → Rar Plains (laterite deposits)

Q: Mithila Plains — between which rivers? → Gandak and Kosi
 Q: Doab meaning → land between two rivers ('do' = two, 'ab' = water)
 Q: Chhosis — what are they? → Gully erosion in Punjab plains caused by seasonal streams

Brahmaputra Plains (Assam)

Brahmaputra carries the largest amount of sediment in the world. Due to high sediment load it forms **braided channels** and creates river islands like **Majuli** — world's largest river island district in Assam.

EXAM FOCUS

Brahmaputra = world's largest sediment-carrying river
 Majuli = largest river island district in India (Assam)
 In Tibet: Brahmaputra = Yarlung Tsangpo | In Arunachal: Dihang | In Assam: Brahmaputra

3.3 Peninsular Plateau

Key Features

Oldest and most stable landmass of India — formed ~2.5 billion years ago (Archaean era)
 Was originally part of **Gondwana Land** (before continental drift)
 Formed by junction of ancient **Cratons** (Aravalli Craton, Singhbhum Craton, Dharwar Craton)
 Average elevation: **600 m** above sea level
 Covers ~16 lakh sq km (about half of India)
 No volcanic activity; tectonically stable; composed of ancient crystalline/metamorphic rocks

Major Hills and Ranges of Peninsular India

Range	Location	Highest Peak / Notes
Aravalli	Rajasthan–Gujarat–Delhi	Guru Shikhar (1722m) in Mt Abu; oldest range in India
Vindhya	MP — N boundary of Deccan	Separates Ganga basin from Deccan plateau
Satpura	MP — between Narmada & Tapi	Pachmarhi (highest point in MP); Dhupgarh peak
Western Ghats	Kerala-Karnataka-Goa-Maharashtra coast	Anaimudi (2695m) — highest peak in peninsular India
Eastern Ghats	AP-TN-Odisha coast	Jindhagada Peak (highest); discontinuous; lower than W Ghats
Nilgiri Hills	Where W & E Ghats meet (TN-Kerala)	Ooty (Udhagamandalam); Doddabetta peak (2637m)
Anaimalai Hills	TN-Kerala boundary	Anaimudi peak
Cardamom Hills	Kerala (Idukki)	Source of many Kerala rivers
Shevroy / Javadi Hills	TN	Part of Eastern Ghats system

★ Frequently Asked in Exams

Aravalli = Oldest mountain range in India (and world — Precambrian)
 Anaimudi (2695m) = Highest peak of Peninsular India / South India (Western Ghats)
 Guru Shikhar (1722m) = Highest peak of Aravalli / Rajasthan
 Western Ghats = UNESCO World Heritage Site (biodiversity hotspot)

Eastern Ghats are discontinuous (rivers cut through them flowing to Bay of Bengal)

Major Plateaus of India

Plateau	Location	Importance
Malwa Plateau	MP + Rajasthan	Between Aravalli and Vindhya; Chambal flows through
Chhota Nagpur Plateau	Jharkhand	Rich in minerals; coal, iron ore; Damodar valley coalfields
Deccan Plateau	Maharashtra, AP, Karnataka	Largest plateau in India; black soil (regur); Deccan Lava flows
Meghalaya Plateau	Meghalaya	Extension of Chhota Nagpur; separated by Brahmaputra plains; wettest place (Mawsynram/Cherrapunji)
Mysore Plateau	Karnataka	Dense forests; coffee and silk production
Telangana Plateau	Telangana	Deccan Trap basalts; Krishna-Godavari basin
Bundelkhand Plateau	UP + MP	Ravines and gullies; Chambal badlands
Baghelkhand Plateau	MP + CG	Coal deposits

🔥 EXAM FOCUS

Q: Which is India's largest plateau? → Deccan Plateau

Q: Mineral-rich plateau → Chhota Nagpur (called 'Ruhr of India')

Q: Wettest plateau region → Meghalaya (Mawsynram/Cherrapunji)

Q: Black cotton soil associated plateau → Deccan (from Deccan Trap basalt weathering)

3.4 Thar Desert (Great Indian Desert)

Aspect	Details
Location	Mostly Rajasthan; parts of Gujarat, Punjab, Haryana
Area	~2 lakh sq km
Annual Rainfall	<25 cm (arid zone)
Saline Lakes	Sambhar, Didwana, Kuchaman, Pachpadra, Degana (salt production)
Crops grown	Bajra, jowar, ragi (millets); pulses
Energy potential	Solar & wind energy hub
Mineral resources	Barmer & Jaisalmer — oil and natural gas fields
Region name	Semi-arid Rajasthan → 'Rajasthan Bangar'; Aravalli = semi-arid boundary

★ Frequently Asked in Exams

Sambhar = largest inland saltwater lake in India (Rajasthan)

Luni River originates from Aravalli → flows to Rann of Kutch (Gujarat)

Barchans = crescent-shaped sand dunes (wind direction indicator)

UPSC has asked about saline lakes of Rajasthan — Sambhar, Didwana, Kuchaman, Pachpadra

3.5 Coastal Plains

Coast	Location	Features
Malabar Coast (West)	Kerala	Backwaters (Kayal); coconut; fishing; Vembanad Lake; Periyar; Pamba rivers
Konkan Coast (West)	Goa + S Maharashtra	Rocky; narrow; Goa beaches; Mandavi, Zuari rivers
Karnataka Coast (West)	Karnataka	Mangalore port; cashew; Netravati river
Gujarat Coast (West)	Gujarat	Gulf of Khambhat; Gulf of Kutch; Sabarmati; Mahi; Luni rivers
Coromandel Coast (East)	TN + South AP	Bay of Bengal; receives NE monsoon rains Oct-Dec; Palk Bay
Northern Circars (East)	North AP + Odisha	Chilika Lake; deltas of Mahanadi, Godavari, Krishna, Kaveri

3.6 Islands of India

Andaman and Nicobar Islands (Bay of Bengal)

Feature	Details
Total Islands	572+ (only 38 inhabited)
Groups	Andaman (North) + Nicobar (South) — separated by 10° Channel
Capital	Port Blair (South Andaman)
Highest Peak	Saddle Peak (North Andaman) — ~732 m
Active Volcano	Barren Island (only active volcano in South Asia)
Dormant Volcano	Narcondam Island
Renamed Islands	Ross Island → Netaji Subhash Chandra Bose Island; Havelock → Swaraj Dweep; Neil Island → Shaheed Dweep
Formation	Submerged mountains (extension of Himalayan arc)

Channels between Andaman-Nicobar Islands

Channel	Separates
10° Channel	Little Andaman (S) from Car Nicobar (N) — most important
Duncan Passage	South Andaman and Little Andaman
St. George's Channel	Little Nicobar and Great Nicobar
Grand Channel	Great Nicobar and Sumatra (Indonesia)

Lakshadweep Islands (Arabian Sea)

Feature	Details
Total Islands	36 islands
Capital	Kavaratti

Feature	Details
Type	Coral islands (Atolls) — formed on coral reefs
Smallest UT	Smallest Union Territory of India by area (~32 sq km)
Groups	Amindivi Islands + Laccadive Islands + Minicoy Island

Important Channels — Lakshadweep

Channel	Separates
9° Channel	Lakshadweep (main) and Minicoy Island
8° Channel	Minicoy Island and Maldives
11° Channel	Amindivi Islands and main Lakshadweep group

EXAM FOCUS

Barren Island = India's only active volcano (Andaman & Nicobar)

Narcondam = Dormant volcano (Andaman & Nicobar)

10° Channel = between Andaman and Nicobar groups

9° Channel = separates main Lakshadweep from Minicoy

Lakshadweep = smallest UT | Andaman & Nicobar = largest UT

Lakshadweep = Coral | Andaman = Submerged mountains (volcanic/sedimentary rock)

CHAPTER 4: INDIAN MONSOON

4.1 Origin of the Word 'Monsoon'

The word '**Monsoon**' is derived from the Arabic word '**Mausam**' (meaning 'season'). Ancient Arab traders navigating to India noticed seasonal reversal of winds — SW winds in summer (to India) and NE winds in winter (back to Arabia). This seasonal wind reversal became associated with 'mausam'.

4.2 Characteristics of Indian Monsoon

Characteristic	Description
Onset	Sudden — arrives over Kerala around June 1 (official onset date)
Retreat	Gradual — takes weeks to retreat; not sudden like onset
Break in Monsoon	1–2 week dry spells within monsoon season (July–August)
Variability	Rainfall amount varies year to year (may fail or be excess)
Duration	June to September (SW Monsoon); Oct–Dec in Tamil Nadu (NE Monsoon)

4.3 Factors Controlling Indian Monsoon (Modern Theory)

1. Tibetan Plateau

Tibet's high altitude causes intense **insolation** (solar heating) → generates an **intense low pressure** over Tibet. This acts as:

- A **source of low pressure** → attracts moisture-laden winds from Indian Ocean
- A **mechanical barrier** → deflects winds southward into the Indian subcontinent

2. Jet Streams

Jet Stream	Season	Effect
Sub-tropical Westerly Jet Stream	Winter	Flows S of Himalayas → brings Western Disturbances → winter rains in NW India
Tibetan Anticyclonic Circulation (Upper Easterly Jet)	Summer	Flows over India from E → helps monsoon surge northward

3. ITCZ (Inter-Tropical Convergence Zone)

The ITCZ is a low-pressure belt where NE and SE trade winds converge. In summer, the ITCZ **shifts northward to the sub-continent** (Thar Desert region), creating a powerful low-pressure trough that draws in the moisture-laden SW monsoon winds.

4. ENSO (El Nino Southern Oscillation)

Phenomenon	Effect on Indian Monsoon
El Nino (warm phase)	Weakens Indian monsoon → drought tendency in India
La Nina (cool phase)	Strengthens Indian monsoon → above-normal rainfall
Southern Oscillation	Pressure difference between Darwin (Aus) and Tahiti (Pacific)
IOD — Indian Ocean Dipole	Positive IOD: above-normal monsoon; Negative IOD: below-normal

🔥 EXAM FOCUS

El Nino → drought in India (weakens monsoon)

La Nina → floods in India (strengthens monsoon)

ITCZ shifts north in summer → triggers SW Monsoon

Mascarene High (near Madagascar in Indian Ocean) = source of high pressure driving monsoon winds

SW Monsoon arrives Kerala ~June 1; withdraws from NW India by mid-September

4.4 Branches of SW Monsoon

Branch	Path	States Covered
Arabian Sea Branch	Hits Western Ghats first; heaviest rain on windward side	Kerala, Karnataka, Goa, Maharashtra, Gujarat
Bay of Bengal Branch	Hits Eastern India and curves NW	Odisha, WB, NE India, then UP, Punjab, Rajasthan

Rain Shadow effect: Eastern side of Western Ghats (leeward side) receives very little rain — hence Deccan Plateau is relatively dry.

★ Frequently Asked in Exams

Mawsynram (Meghalaya) = Wettest place in India and world (avg ~11,871 mm/year)

Cherrapunji = Second wettest; historically recorded highest rainfall

Chennai receives most rain from NE Monsoon (Oct–Dec), not SW Monsoon

Jaisalmer = driest place in India (Thar Desert)

CHAPTER 5: SOILS OF INDIA

5.1 Factors Affecting Soil Formation

Factor	Influence
Parent Rock (Geology)	Determines mineral composition of soil
Climate	Controls weathering rate; rainfall → leaching; temperature → decomposition
Topography	Plains = deep homogeneous soil; Mountains = shallow heterogeneous soil
Biotic Factors	Vegetation → humus content; bacteria → decomposition
Time	More time = more mature/deep soil

5.2 Soil Profile (Horizons)

Horizon	Name	Description
O Horizon	Organic Layer	Surface layer; undecomposed/partly decomposed leaf litter, dead matter
A Horizon	Top Soil	Most fertile; all minerals present; humus content here; zone of eluviation (E layer within A)
E Layer	Zone of Eluviation	Within A horizon; minerals leach out downward
B Horizon	Sub Soil	Zone of illuviation; iron, aluminum, humic compounds accumulate here
C Horizon	Parent Material	Partially weathered rock; less altered than above
R Horizon	Bedrock	Unweathered solid rock

5.3 Types of Soils in India (8 types as per ICAR)

Soil Type	Distribution	Characteristics	Key Crops
Alluvial Soil	N Plains (UP, Bihar, Punjab, WB), river deltas	Most extensive; Bangar (old) & Khadar (new); Kankars in Bangar; highly fertile	Wheat, rice, sugarcane, cotton, jute
Black Soil (Regur)	Deccan Plateau (Maharashtra, MP, Gujarat, AP)	Formed from Deccan Lava (basalt); swells when wet, cracks when dry; self-ploughing; high clay content	Cotton (best soil for cotton); soybean
Red & Yellow Soil	Eastern MP, Odisha, S AP, Kerala, Tamil Nadu	Formed on ancient crystalline rocks; red due to iron oxides; poor in N, P, humus	Rice, millet, tobacco, groundnut
Laterite Soil	Kerala, Karnataka, TN, Odisha highlands (humid tropics)	Formed by intense leaching in high rainfall; hard when dry; poor in N, Ca; rich in Fe, Al	Tea, coffee, rubber, cashew
Arid/Desert Soil	Rajasthan, S Punjab, N Gujarat	Sandy; low humus; high salt content; irrigation makes fertile	Bajra, pulses with irrigation
Saline/Alkaline Soil	Arid areas, coastal areas	Reh/Usar/Thur soil; high salt; poor fertility; waterlogging issues	Problem soil — needs reclamation

Soil Type	Distribution	Characteristics	Key Crops
Peaty/Marshy Soil	Kerala, Bengal, Bihar lowlands	High organic/humus content; waterlogged; acidic	Rice, vegetables
Forest/Mountain Soil	Himalayan + Eastern Ghats forests	Varies with altitude; thin; high organic matter	Tea, coffee, spices in lower altitudes

EXAM FOCUS

- Q: Which soil is best for cotton? → Black (Regur) soil
- Q: Which soil covers largest area in India? → Alluvial soil
- Q: Laterite soil is formed by → intense leaching in heavy rainfall areas
- Q: Self-ploughing property → Black soil (swells + cracks)
- Q: Podzolization → process in coniferous forests where sesqui-oxides of Fe & Al leach, leaving silica+calcium
- Q: Laterization → process in high rainfall where SiO₂ and Ca leach, leaving Fe and Al oxides
- Q: Reh/Usar = Saline soil found in Indo-Gangetic Plains

CHAPTER 6: NATURAL VEGETATION OF INDIA

6.1 Forest Cover

According to India State of Forest Report, forest & tree cover = ~24.6% of India's total geographical area. Constitutional provision: Article 51-A (g) → fundamental duty to protect environment/forest/wildlife.

6.2 Types of Forests

Forest Type	Rainfall	Location	Key Trees/Features
Tropical Evergreen	200+ cm; no dry season	N-E India, Western Ghats windward side, A&N Islands	Ebony, mahogany, rosewood; dense canopy; no leaf-shedding season; highest biodiversity
Tropical Deciduous (Moist)	100–200 cm	E MP, Jharkhand, Odisha, Chhattisgarh, N-E India	Teak, sal, sandalwood; shed leaves in dry season; most commercially important
Tropical Deciduous (Dry)	70–100 cm	Central India plains (UP, Bihar, Rajasthan, MP)	Sal, kendu, neem, peepal; more open canopy
Tropical Thorn Scrub	<70 cm	Rajasthan, Gujarat, Punjab, Haryana dry areas	Khejri (state tree of Rajasthan), cacti, babul (Acacia)
Montane Temperate	Altitude-based (900–2700m)	Himalayas, Nilgiris	Oak, chestnut, deodar, pine, rhododendron
Alpine	Above 3600m	High Himalayas	Juniper, silver fir, birch; scrub at snowline
Mangrove / Tidal	Coastal+tidal zone	Sundarbans (WB), Gujarat, Andhra, Odisha	Sundari tree; salt-tolerant; oxygen roots (pneumatophores); habitat for tigers, crocodiles
Littoral / Swamp	Coastal areas	Along sea coasts, lagoons	Coconut, pandanus; brackish water species

★ Frequently Asked in Exams

Sundarbans = world's largest mangrove forest (India + Bangladesh); UNESCO WH Site

Sundari tree → Sundarbans name derived from it

Western Ghats = biodiversity hotspot (one of 36 globally)

India has 18 biosphere reserves recognized by UNESCO Man and Biosphere Programme

Largest forest area by state: Madhya Pradesh

Highest % forest cover: Mizoram (~85%)

CHAPTER 7: DRAINAGE SYSTEM OF INDIA

7.1 Classification of Indian Rivers

Basis	Types
By Origin	Himalayan Rivers (Perennial — snow/glacier fed) vs Peninsular Rivers (Seasonal — rain-fed)
By Drainage Basin	Bay of Bengal drainage vs Arabian Sea drainage vs Inland drainage
By Flow	Antecedent (pre-dates mountain — cuts through) vs Superimposed (overlaid on rock) vs Consequent (follows slope)
By Pattern	Dendritic, Radial, Centripetal, Trellis, Parallel

Drainage Patterns

Pattern	Description	Example
Dendritic	Tree-like branching; on uniform rock	Ganga, Indus
Radial	Rivers flow outward from a central hill	Amarkantak hills → Narmada, Son, Mahanadi radiate outward
Centripetal	Rivers flow inward to a depression/lake	Rivers of Rann of Kutch, Loktak Lake basin
Trellis	Tributaries join at right angles	Rivers of folded Appalachian type
Angular	Between rectangular and trellis	River systems in jointed rocks

7.2 Major Himalayan Rivers

Indus River System

Feature	Details
Origin	Bokhar Chu Glacier near Kailash Mansarovar Lake (Tibet)
Length	~3180 km (total); ~1114 km in India
Flow direction	Flows NW through Ladakh → enters Pakistan; meets Arabian Sea near Karachi
Left bank tributaries (in India)	Zaskar, Shyok, Nubra, Gilgit (in Ladakh)
Right bank (Punjab 5 rivers)	Jhelum, Chenab, Ravi, Beas, Sutlej
Treaty	Indus Waters Treaty 1960 (India-Pakistan mediated by World Bank)
Gorge formed	Indus Gorge — deepest gorge in the world between Ladakh and Gilgit

★ Frequently Asked in Exams

Indus Waters Treaty (1960): India gets E rivers (Sutlej, Beas, Ravi); Pakistan gets W rivers (Indus, Jhelum, Chenab)

Indus Gorge = world's deepest gorge

Sutlej enters India via Shipki La Pass

Satluj + Beas join = forms Sutlej in Pakistan → then joins Indus

Ranjit Sagar Dam = on Ravi River (J&K)

Ganga River System

Feature	Details
Origin	Gangotri Glacier (Bhagirathi + Alaknanda join at Devprayag = Ganga)
Length	~2525 km in India; largest by discharge in India
States covered	Uttarakhand, UP, Bihar, Jharkhand, WB
Major Left bank tributaries	Ramganga, Gomti, Ghaghra (Karnali), Gandak, Kosi, Mahananda
Major Right bank tributaries	Yamuna, Son, Damodar (joins Hooghly)
National River	Ganga declared national river in 2008
Key tributaries' origins	Yamuna: Yamunotri glacier; Kosi: Nepal Himalayas (Sorrow of Bihar); Son: Amarkantak

Prayagraj Sangam: Ganga, Yamuna, and mythological Saraswati meet here — site of Kumbh Mela.

River	Origin	Notable Features
Yamuna	Yamunotri Glacier (Uttarakhand)	Longest tributary of Ganga; Delhi, Agra, Mathura on banks; joins at Prayagraj
Ghaghra/Karnali	Nepal Himalayas	Meets Ganga near Chapra (Bihar); highest discharge tributary
Gandak/Narayani	Nepal	Joins Ganga near Patna
Kosi	Nepal (meets 5 rivers)	'Sorrow of Bihar' — changes course frequently; flood-prone
Chambal	Vindhya/Janapav Hills (MP)	Flows through ravines; tributes to Yamuna; Chambal WL Sanctuary; Gandhi Sagar Dam
Son	Amarkantak Plateau	Right bank tributary; joins near Patna
Ramganga	Nainital Hills (Uttarakhand)	Right bank; Rohelkhand Plains formed by its alluvium

🔥 EXAM FOCUS

Ganga = National River | Gangetic Dolphin = National Aquatic Animal
 Haridwar = where Ganga enters plains | Rishikesh = rapids for rafting
 Kosi = 'Sorrow of Bihar' — deposits sediment + changes course = floods
 Yamuna = longest tributary of Ganga
 Son River forms Magadh Plains (south of Son)
 Chambal = originates from Vindhyas (Janapav Hill, MP) → ravines → Yamuna

Brahmaputra River System

Feature	Details
Origin	Chemayungdung (Chema Yungdung) Glacier, Tibet
Names by region	Tibet: Yarlung Tsangpo → Arunachal: Dihang (Siang) → Assam: Brahmaputra → Bangladesh: Jamuna
Total length	~2900 km (one of the longest rivers in the world)

Feature	Details
Key feature	Antecedent river; older than Himalayas; syntaxial bend at Namcha Barwa
Sediment	World's largest sediment-carrying river
Major island	Majuli (world's largest river island district — Assam)
Left bank tributaries	Lohit, Dibang, Dihang, Dhansiri
Right bank tributaries	Subansiri, Kameng, Manas, Sankosh, Teesta
Bangladesh	Joins Ganga (as Padma) + Meghna → Bay of Bengal; world's largest delta (Sundarbans)

🔥 EXAM FOCUS

Brahmaputra = largest river by water volume/discharge in India

Namcha Barwa = syntaxial bend — eastern most point of Himalayas

Teesta River → tributary of Brahmaputra; dispute between India-Bangladesh-Sikkim

Manas River → important tributary; Manas National Park on its banks (Assam/Bhutan border)

Tista = Teesta; important for West Bengal & Sikkim

7.3 Peninsular Rivers

East-Flowing Peninsular Rivers

River	Origin	Tributaries	Mouth/States
Mahanadi	Raipur district, Chhattisgarh	Seonath, Hasdeo, Mand, Ib (left); Jonk, Ong, Tel (right)	Bay of Bengal (Odisha); Hirakud Dam
Godavari	Trimbakeshwar (Nasik, Maharashtra)	Indravati, Pranhita (Ward+Wainganga+Penganga), Sabari	Bay of Bengal (AP); 'Dakshin Ganga'
Krishna	Mahabaleshwar (Maharashtra)	Ghataprabha, Malaprabha, Koyna, Bhima, Tungabhadra	Bay of Bengal (AP/Telangana); Nagarjuna Sagar Dam
Kaveri (Cauvery)	Brahmagiri Hills (Coorg, Karnataka)	Hemavati, Arkavati, Shimsha (left); Kabini, Bhavani, Amravati (right)	Bay of Bengal (TN); KRS Dam; Cauvery water dispute
Suvarnarekha	Ranchi Plateau (Jharkhand)	—	Bay of Bengal (Odisha/WB)
Brahmani	Junction of Sankh+South Koel (Jharkhand)	—	Bay of Bengal (Odisha)
Baitarni	Eastern Ghats (Odisha)	—	Bay of Bengal (Odisha)

★ Frequently Asked in Exams

Godavari = Largest peninsular river | 'Dakshin Ganga' (Ganges of South)

Hirakud Dam (Odisha) = longest dam in India — on Mahanadi

Nagarjuna Sagar Dam = on Krishna River (Telangana/AP)

KRS Dam (Krishnarajasagar) = on Kaveri River (Karnataka) — Cauvery dispute

Tungabhadra (trib. of Krishna) → Hampi is on its banks
 Shivsamudram Waterfalls = on Kaveri River (Karnataka)
 Indravati Nat. Park = on Indravati River (Chhattisgarh); Project Tiger
 Kolleru Lake (AP) = freshwater lake formed at delta of Godavari+Krishna

West-Flowing Peninsular Rivers

River	Origin	Flow / Features	Mouth
Narmada	Amarkantak Plateau (MP)	Rift valley river (between Vindhya & Satpura); Dhuandhar Falls; Sardar Sarovar Dam; Indira Sagar Dam; Tawa (L.bank), Kolar (R.bank) tributaries	Gulf of Khambhat (Arabian Sea)
Tapti (Tapi)	Betul Plateau (MP)	Also rift valley (S of Satpura); parallel to Narmada	Gulf of Khambhat (Arabian Sea)
Luni	Aravalli (near Ajmer, Rajasthan)	Only river of Rajasthan that reaches sea; saline after Balotra	Rann of Kutch
Sabarmati	Udaipur (Aravalli, Rajasthan)	Ahmedabad, Gandhi Ashram on banks; Dharoi Dam	Gulf of Khambhat
Mahi	Vindhya Range (MP)	Crosses Tropic of Cancer twice; Kadana Dam	Gulf of Khambhat
Periyar	Cardamom Hills (Kerala)	Longest river of Kerala; Periyar NP (Tiger Reserve)	Arabian Sea
Bharathapuzha	Anaimalai Hills (Kerala)	2nd longest river of Kerala	Arabian Sea
Pamba	Kerala Highlands	Pamba falls into Vembanad Lake; Sabarimala Temple on banks	Vembanad Lake → Arabian Sea
Mandavi / Mahadayi	Western Ghats (Goa/Karnataka)	Panaji is on Mandavi River; casino boats here	Arabian Sea (Goa)
Zuari	Western Ghats (Goa)	Meets Mandavi; Old Goa between two rivers	Arabian Sea (Goa)
Sharavati	Sahyadri/Western Ghats (Karnataka)	Jog Falls (Gersoppa Falls) on Sharavati — highest plunge waterfall in India	Arabian Sea

🔥 EXAM FOCUS

Narmada & Tapti = west-flowing rift valley rivers (flow in grabens between parallel faults)
 Narmada: Sardar Sarovar Dam (Gujarat) + Indira Sagar Dam (MP)
 Mahi crosses Tropic of Cancer TWICE — unique feature
 Jog Falls (Gersoppa Falls) = on Sharavati River, Karnataka = highest plunge waterfall (~253m)
 Pamba River → Sabarimala Temple (in news for Supreme Court judgment)
 Luni River = only river of Rajasthan; becomes saline/brackish after Balotra
 Sabarmati Ashram (Gandhi) = on Sabarmati River, Ahmedabad
 Zuari + Mandavi = rivers of Goa

CHAPTER 8: IMPORTANT LAKES OF INDIA

Lake	State	Type	River/Notes
Chilika Lake	Odisha	Brackish (Lagoon); largest brackish lake in India; Ramsar site	Bay of Bengal coast; flamingos; Irrawaddy dolphin
Vembanad Lake	Kerala	Brackish (Backwater); longest lake in India (~96km)	Pamba, Meenachil rivers; Kuttanad (below sea level farming)
Kolleru Lake	Andhra Pradesh	Freshwater; deltaic lake of Godavari+Krishna	Natural freshwater lake; Ramsar site
Pulicat Lake	Andhra Pradesh / Tamil Nadu	Brackish; 2nd largest saltwater lake	Flamingo Festival; Sriharikota (ISRO) is on a sandbar
Loktak Lake	Manipur	Freshwater + Phumdis (floating mass of vegetation)	Keibul Lamjao NP (floating NP); Sangai Deer
Wular Lake	J&K	Freshwater; largest freshwater lake in India	On Jhelum River
Dal Lake	J&K	Freshwater; famous houseboat lake	On Jhelum tributary; tourism
Pangong Tso	Ladakh (India+China)	Saline; changes color (blue-green); 60% in China	At ~4350m altitude; India-China LAC passes through it
Tso Moriri	Ladakh	High altitude lake	Ramsar site; wetland conservation
Sambhar Lake	Rajasthan	Saltwater; largest inland salt lake in India	Salt production; Ramsar site
Pushkar Lake	Rajasthan	Freshwater	Sacred; Brahma Temple; Camel Fair
Pichola Lake	Rajasthan (Udaipur)	Artificial	Famous tourist lake; Lake Palace hotel
Bhoj Tal (Upper/Lower Lake)	MP (Bhopal)	Artificial	Bhopal = 'City of Lakes'
Hussain Sagar	Telangana (Hyderabad)	Artificial	On Musi River tributary; Buddha statue in center
Govind Sagar	Himachal Pradesh	Artificial reservoir	Bhakra Nangal Dam on Sutlej River
Lonar Lake	Maharashtra	Saline + Alkaline; impact crater lake	Only hypervelocity impact crater in basaltic rock; Ramsar site
Shivajisagar / Koyna Reservoir	Maharashtra	Artificial	Koyna Dam on Krishna tributary
Deepor Beel	Assam	Freshwater; wetland	On Brahmaputra; Ramsar site; elephant habitat
Renuka Lake	Himachal Pradesh	Natural; famous pilgrimage	Largest natural lake in HP

Lake	State	Type	River/Notes
Sattal Lake	Uttarakhand (Nainital)	Freshwater; cluster of 7 lakes	Migratory birds
Kanwar Lake	Bihar	Oxbow lake on Gandak River	Wetland; water birds

EXAM FOCUS

Wular Lake = largest freshwater lake in India (J&K, on Jhelum)

Chilika Lake = largest brackish water / coastal lagoon in India (Odisha)

Sambhar = largest inland saltwater lake in India (Rajasthan)

Vembanad = longest lake in India by length (~96 km)

Lonar Lake = only meteorite impact crater lake in basaltic rock (Maharashtra)

Keibul Lamjao NP = world's only floating National Park (on Loktak Lake, Manipur)

Phumdis = floating mass of soil, vegetation on Loktak Lake

Pangong Tso = 60% in China (in news during India-China Galwan crisis)

Kolleru + Sambhar + Chilika = Ramsar wetland sites

CHAPTER 9: MAJOR DAMS OF INDIA

Dam	River	State	Notable Feature
Tehri Dam	Bhagirathi (Ganga)	Uttarakhand	Highest dam in India (~261m); largest reservoir
Bhakra Nangal	Sutlej	HP/Punjab border	Tallest gravity dam in Asia; Gobind Sagar reservoir
Hirakud	Mahanadi	Odisha	Longest dam in India (~4.8 km)
Sardar Sarovar	Narmada	Gujarat	Controversial; Narmada Bachao Andolan; 2nd largest in India
Indira Sagar	Narmada	MP	Largest reservoir in India by storage
Nagarjuna Sagar	Krishna	Telangana/AP	One of world's largest masonry dams
Krishna Raja Sagar (KRS)	Kaveri	Karnataka	Visvesvaraya Dam; water to Bengaluru/Mysuru
Idukki	Periyar	Kerala	Arch dam (largest in Asia); hydropower
Gandhi Sagar	Chambal	MP	Chambal Valley Project
Jawahar Sagar	Chambal	Rajasthan	Part of Chambal Valley Project
Rana Pratap Sagar	Chambal	Rajasthan	—
Tawa Dam	Tawa (trib. of Narmada)	MP	—
Koyna Dam	Koyna (trib. of Krishna)	Maharashtra	Reservoir: Shivajisagar; caused earthquake 1967
Tungabhadra Dam	Tungabhadra	Karnataka	Hampi area
Mettur Dam	Kaveri	Tamil Nadu	Oldest dam in South India
Rihand Dam	Rihand (trib. of Son)	UP	Govind Ballabh Pant Sagar — largest man-made lake (volume) in India
Nathpa Jhakri	Sutlej	HP	Largest hydroelectric project in India
Ranjit Sagar	Ravi	J&K	Thein Dam
Dulhasti	Chenab	J&K	—
Baglihar	Chenab	J&K	—

★ Frequently Asked in Exams

Tehri = Tallest dam in India (~261m) | Bhakra = Tallest GRAVITY dam

Hirakud = Longest dam in India (~4.8 km)

Indira Sagar = Largest reservoir in India (by storage capacity)

Govind Ballabh Pant Sagar (Rihand Dam) = Largest man-made lake in India by volume/area

Idukki = Largest arch dam in Asia (Kerala)

Koyna earthquake 1967 = Reservoir-induced seismicity

All 3 Chambal dams (Gandhi Sagar + Rana Pratap + Jawahar Sagar) are in MP/Rajasthan

CHAPTER 10: NATIONAL PARKS, WILDLIFE & BIOSPHERE RESERVES

10.1 Major National Parks of India

National Park	State	Famous For
Jim Corbett NP	Uttarakhand	India's oldest NP (1936); Bengal Tiger; Project Tiger (1st)
Kaziranga NP	Assam	One-horned Rhinoceros (largest population); UNESCO WH; Brahmaputra
Sundarban NP	West Bengal	Royal Bengal Tiger; Mangroves; UNESCO WH; Gangetic Delta
Gir Forest NP	Gujarat	Only habitat of Asiatic Lion in wild; Sasan Gir
Keibul Lamjao NP	Manipur	World's only floating NP; Sangai/Eld's Deer; on Loktak Lake
Hemis NP	Ladakh (J&K)	Snow Leopard; largest NP in India
Periyar NP	Kerala	Elephants; Tigers; Periyar River; TR
Bandipur NP	Karnataka	Tigers; Elephants; adjoins Mudumalai & Nagarhole
Nagarhole NP	Karnataka	Elephants; Kabini River; Rajiv Gandhi NP
Pench NP	MP/Maharashtra border	Tigers; Kipling's Jungle Book setting
Kanha NP	MP	Tigers; Barasingha (hardground deer)
Bandhavgarh NP	MP	Highest tiger density in India
Panna NP	MP	Tigers; Ken River; TR
Nauradehi WS (MP)	MP	India's proposed cheetah reintroduction site
Sariska NP	Rajasthan	Tigers; Alwar district; TR
Ranthambore NP	Rajasthan	Tigers; Ranthambore Fort; TR
Keoladeo Ghana NP	Rajasthan	Siberian crane; migratory birds; UNESCO WH; Bharatpur Bird Sanctuary
Manas NP	Assam	Tiger + Rhinoceros + Elephant; UNESCO WH; Project Tiger; on Bhutan border
Indravati NP	Chhattisgarh	Wild Buffalo; on Indravati River; TR
Simlipal NP	Odisha	Tigers; Elephants; Barehipani Falls
Eravikulam NP	Kerala	Nilgiri Tahr; Anamudi peak
Silent Valley NP	Kerala	Lion-tailed Macaque; no roads inside
Great Himalayan NP	HP	UNESCO WH; Snow Leopard; Western Himalayan biodiversity
Dachigam NP	J&K	Hangul (Kashmir Stag)
Bharatpur Bird Sanctuary	Rajasthan	Migratory birds; UNESCO WH
Fangpui (Phawngpui) Blue Mountain NP	Mizoram	Blue rocks; highest peak of Mizoram

🔥 EXAM FOCUS

Jim Corbett = India's FIRST National Park (1936); 1st Project Tiger park (1973)
 Kaziranga = largest population of one-horned rhinos in the world
 Gir = only habitat of Asiatic Lion
 Keibul Lamjao = world's only floating National Park
 Hemis = largest National Park in India (Ladakh)
 Keoladeo Ghana = UNESCO WH; Siberian Crane migration destination
 Manas NP spans India-Bhutan border (Royal Manas NP in Bhutan)
 Sundarbans = largest mangrove + Bengal Tiger habitat; UNESCO WH
 Simlipal NP = India's 2nd largest biosphere reserve

10.2 Tiger Reserves (Project Tiger — launched 1973)

India has the **highest number of wild tigers** in the world (>3,000 as of 2022). Project Tiger was launched in 1973 under PM Indira Gandhi. NTCA (National Tiger Conservation Authority) manages tiger reserves.

States with most Tiger Reserves: **Madhya Pradesh** → 'Tiger State of India'

★ Frequently Asked in Exams

India has 54 Tiger Reserves (as of 2023)
 First Tiger Reserve = Jim Corbett (1973)
 Maximum tigers = MP (Madhya Pradesh) — 'Tiger State'
 Maharashtra's Umred-Pauni-Karhandla = Jai the Tiger
 Orang NP (Assam) = Mini Kaziranga — rhinos + tigers

10.3 UNESCO Biosphere Reserves in India

Biosphere Reserve	State
Sundarbans	West Bengal
Gulf of Mannar	Tamil Nadu
Nilgiris	TN + Kerala + Karnataka
Nokrek	Meghalaya
Nanda Devi	Uttarakhand
Manas	Assam
Simlipal	Odisha
Dibru-Saikhowa	Assam
Dihang-Dibang	Arunachal Pradesh
Khangchendzonga	Sikkim
Panna	MP
Achanakmar-Amarkantak	MP + CG
Agasthyamalai	Kerala + TN
Great Nicobar	Andaman & Nicobar
Pachmarhi	MP

Biosphere Reserve	State
Kanha-Pench	MP

10.4 Ramsar Wetland Sites

India has **75+ Ramsar Sites** (as of 2023) — highest number in Asia. Key sites:

Ramsar Site	State	Notable Feature
Chilika Lake	Odisha	Largest coastal wetland; Irrawaddy dolphin; flamingo
Keoladeo Ghana	Rajasthan	Migratory birds; Siberian crane
Sambhar Lake	Rajasthan	Salt production; flamingo
Vembanad-Kol	Kerala	Largest wetland in India; backwaters
Wular Lake	J&K	Largest freshwater lake
Loktak Lake	Manipur	Floating phumdis
Harike Wetland	Punjab	Junction of Sutlej and Beas
Deepor Beel	Assam	Freshwater lake; elephants
Tso Moriri	Ladakh	High altitude wetland; migratory birds
Lonar Lake	Maharashtra	Unique meteor crater lake; saline + alkaline
Renuka Lake	Himachal	Smallest Ramsar site in India
Nalsarovar	Gujarat	Migratory birds; flamingo

EXAM FOCUS

India = Most Ramsar wetland sites in Asia

1st Ramsar sites in India: Chilika (Odisha) & Keoladeo (Rajasthan) — 1981

Ramsar Convention (1971) — signed at Ramsar, Iran

Renuka Lake (HP) = smallest Ramsar site in India

Vembanad-Kol (Kerala) = largest Ramsar site in India

Lonar Lake added 2020 — unique alkaline + saline + meteor crater

CHAPTER 11: POPULATION & SETTLEMENTS (OVERVIEW)

Aspect	Data / Facts
Census 2011 Population	121.07 crore (1.21 billion)
Decadal Growth Rate (2001-11)	17.64% (declined from 21.54% in 1991-2001)
Density (2011)	382 persons per sq km
Most populous state	UP (Uttar Pradesh)
Least populous state	Sikkim
Highest density	Bihar (1106/sq km)
Lowest density	Arunachal Pradesh (17/sq km)
Most literate state	Kerala (94%)
Least literate	Bihar (~61%)
Sex Ratio (2011)	940 females per 1000 males
Best sex ratio	Kerala (1084)
Worst sex ratio	Haryana (879)
Urban Population	31.16% (as per 2011)
India's population share (world)	~18% of world in ~2.4% of land

★ Frequently Asked in Exams

UPSC/UPPCS frequently asks: most populous, most dense, least dense states
 India became world's most populous country (surpassing China) in 2023
 UP = most populous state; Bihar = highest density; Arunachal = lowest density
 Kerala = highest literacy + best sex ratio + highest life expectancy
 As per India Census: rural population still ~69% despite urbanization

CHAPTER 12: WATERFALLS & OTHER IMPORTANT FACTS

12.1 Major Waterfalls

Waterfall	River	State	Height	Notes
Jog Falls (Gersoppa)	Sharavati	Karnataka	~253m	Highest plunge waterfall in India
Nohkalikai Falls	Near Cherrapunji	Meghalaya	~340m	India's tallest waterfall (free fall)
Dhuandhar Falls	Narmada	MP	~30m	'Smoke cascade'; near Bhedaghat, Jabalpur
Dudhsagar Falls	Mandavi	Goa/Karnataka	~310m	Milky white; on Konkan Railway line
Chitrakote Falls	Indravati	Chhattisgarh	~90m	Widest waterfall in India; 'India's Niagara'
Sivasamudram Falls	Kaveri	Karnataka	~98m	Hydropower; Bluff Island
Athirappilly Falls	Chalakudy	Kerala	~80m	Niagara of India (Kerala)
Barehipani Falls	Budhabalanga	Odisha	~399m	2nd highest waterfall; in Simlipal NP
Thalaiyar Falls	Manjalar	Tamil Nadu	~297m	Rat Tail Falls; deepest gorge in India

EXAM FOCUS

Jog/Gersoppa = Highest PLUNGE waterfall in India (Sharavati, Karnataka)
 Nohkalikai = Highest TOTAL fall in India (Meghalaya, near Cherrapunji)
 Dhuandhar Falls = Narmada River, MP (Bhedaghat — Jabalpur)
 Chitrakote Falls = Widest waterfall in India (CG) — 'India's Niagara'
 Dudhsagar = can be seen from Konkan Railway trains (Goa/Karnataka border)

12.2 India's Straits and Channels

Strait / Channel	Connects / Separates	Notes
Palk Strait	India and Sri Lanka	Between Gulf of Mannar and Bay of Bengal; Adam's Bridge (Ram Setu)
10° Channel	Little Andaman and Car Nicobar	Main division between Andaman and Nicobar groups
Duncan Passage	South Andaman and Little Andaman	
9° Channel	Lakshadweep & Minicoy	
8° Channel	Minicoy & Maldives	
Grand Channel	Great Nicobar & Sumatra	India's southernmost point near here (Indira Point)

12.3 India's Extreme Points

Direction	Point	Location
Northernmost	Indira Col	Siachen Glacier, J&K
Southernmost (mainland)	Cape Comorin (Kanyakumari)	Tamil Nadu — confluence of 3 seas
Southernmost (India incl. islands)	Indira Point / Pygmalion Point	Great Nicobar Island (submerged in 2004 Tsunami)
Easternmost	Kibithu (Kibitu)	Arunachal Pradesh
Westernmost	Sir Creek / Guhar Moti	Gujarat (Rann of Kutch)

EXAM FOCUS

Kanyakumari = southernmost tip of mainland India; 3 seas (Arabian Sea, Bay of Bengal, Indian Ocean) meet

Indira Point = India's southernmost point (Great Nicobar) — submerged 2004 Tsunami

Kibithu = easternmost point of India (Arunachal Pradesh)

Indira Col (or Dafadar) = northernmost point (disputed because Siachen)

CHAPTER 13: STATES & UNION TERRITORIES — EXAM REFERENCE

State/UT	Capital	Major River	State Animal	State Bird	National Park/WLS
Uttar Pradesh	Lucknow	Ganga, Yamuna, Ghaghra	Swamp Deer (Barasingha)	Sarus Crane	Dudhwa NP
Madhya Pradesh	Bhopal	Narmada, Tapti, Chambal	Barasingha (Hard Ground)	Indian Paradise Flycatcher	Kanha, Bandhavgarh, Panna NP
Maharashtra	Mumbai	Godavari, Krishna, Tapti	Giant Squirrel	Yellow-footed Green Pigeon	Sanjay Gandhi NP
Rajasthan	Jaipur	Chambal, Luni, Banas	Chinkara	Great Indian Bustard	Ranthambore, Sariska NP
Gujarat	Gandhinagar	Sabarmati, Mahi, Narmada	Asiatic Lion	Greater Flamingo	Gir Forest NP
Bihar	Patna	Ganga, Son, Gandak, Kosi	Gaur	House Sparrow	Valmiki NP
West Bengal	Kolkata	Ganga/Hooghly, Teesta	Fishing Cat	White-breasted Kingfisher	Sundarbans NP, Gorumara NP
Assam	Dispur	Brahmaputra, Barak	Indian One-horned Rhino	White-winged Wood Duck	Kaziranga NP, Manas NP
Karnataka	Bengaluru	Krishna, Kaveri, Tungabhadra	Indian Elephant	Indian Roller	Bandipur, Nagarhole NP
Andhra Pradesh	Amaravati	Krishna, Godavari	Blackbuck	Rose-ringed Parakeet	Sri Venkateswara NP
Telangana	Hyderabad	Krishna, Godavari, Musi	Spotted Deer (Chital)	Palapitta (Blue Rock Pigeon)	Mrugavani NP
Tamil Nadu	Chennai	Kaveri, Palar, Vaigai	Nilgiri Tahr	Emerald Dove	Mudumalai NP, Anamalai NP
Kerala	Thiruvananthapuram	Periyar, Pamba, Bharathapuzha	Indian Elephant	Great Hornbill	Periyar NP, Silent Valley NP
Odisha	Bhubaneswar	Mahanadi, Brahmani, Baitarni	Sambar Deer	Indian Roller (Neelakantha)	Simlipal NP, Bhitarkanika NP
Jharkhand	Ranchi	Damodar, Subanarekha	Indian Elephant	Koel (Asian Koel)	Betla NP
Chhattisgarh	Raipur	Mahanadi, Indravati, Sheonath	Wild Buffalo	Hill Myna	Indravati NP, Kanger Valley NP
Punjab	Chandigarh	Sutlej, Beas, Ravi	Blackbuck	Northern Goshawk	Harike Wetland
Haryana	Chandigarh	Yamuna, Ghaggar	Blackbuck	Black Francolin	Sultanpur Bird Sanctuary

State/UT	Capital	Major River	State Animal	State Bird	National Park/WLS
Himachal Pradesh	Shimla	Sutlej, Ravi, Beas, Chenab	Snow Leopard	Western Tragopan	Great Himalayan NP, Pin Valley NP
Uttarakhand	Dehradun	Ganga (Bhagirathi+Alaknanda), Yamuna	Musk Deer	Himalayan Monal	Jim Corbett NP, Nanda Devi BR
J&K (UT)	Srinagar (S), Jammu (W)	Jhelum, Chenab, Ravi, Indus	Hangul (Kashmir Stag)	Black-necked Crane	Dachigam NP, Hemis NP (Ladakh)
Sikkim	Gangtok	Teesta, Rangit	Red Panda	Blood Pheasant	Khangchendzonga NP
Meghalaya	Shillong	Umiam, Shillong	Clouded Leopard	Hill Myna	Balpakram NP
Arunachal Pradesh	Itanagar	Brahmaputra (Siang/Dihang), Subansiri	Gayal (Mithun)	Great Hornbill	Namdapha NP
Nagaland	Kohima	Dhansiri, Doyang	Mithun	Blyth's Tragopan	Intanki NP
Manipur	Imphal	Imphal, Barak	Sangai Deer	Hume's Pheasant	Keibul Lamjao NP
Mizoram	Aizawl	Tlawng, Kolodyne	Malayan Giant Squirrel	Mrs Hume's Pheasant	Phawngpui Blue Mountain NP
Tripura	Agartala	Haora, Gumti	Phayre's Langur	Green Imperial Pigeon	Clouded Leopard NP
Goa	Panaji	Mandavi, Zuari	Gaur	Black-crested Bulbul	Bhagwan Mahavir WS
Lakshadweep (UT)	Kavaratti	—	Butterfly Fish	Noddy Tern	—
A&N Islands (UT)	Port Blair	—	Dugong (Sea Cow)	Andaman Wood Pigeon	Campbell Bay NP, Galathea NP
Delhi (UT)	New Delhi	Yamuna	Nilgai	House Sparrow	Asola Bhatti Wildlife Sanctuary
Chandigarh (UT)	Chandigarh	—	Indian Grey Mongoose	Indian Grey Hornbill	—

EXAM FOCUS

Chandigarh = shared capital of Punjab and Haryana
 Dehradun = capital of Uttarakhand (shifted from Uttarkashi)
 MP and CG share a border; CG was carved out of MP in 2000
 Jharkhand = carved out of Bihar in 2000 | Uttarakhand = from UP in 2000
 Goa = smallest state by area | Sikkim = least population
 UP = largest state by population | Rajasthan = largest state by area

CHAPTER 14: ENVIRONMENT — LAWS, CONVENTIONS & GOVERNMENT SCHEMES

14.1 Constitutional Provisions on Environment

Article	Provision
Art. 48-A (DPSP)	State shall endeavour to protect and improve environment and safeguard forests and wildlife
Art. 51-A(g) (FD)	Fundamental Duty of citizens to protect environment, forests, rivers, wildlife
Art. 21	Right to Life includes right to live in clean/healthy environment (SC expanded)
Art. 246, 7th Schedule	Environment is in Concurrent List (both Centre and States can legislate)
100th Amendment (2015)	Land Boundary Agreement with Bangladesh — exchange of enclaves

14.2 Important Environmental Laws

Law / Act	Year	Key Provision
Wildlife Protection Act	1972	Protection of wild animals, birds, plants; schedules I-VI; NBWL
Water (Prevention & Control of Pollution) Act	1974	CPCB and SPCBs established
Forest (Conservation) Act	1980	Restrictions on diversion of forest land for non-forest use
Air (Prevention & Control of Pollution) Act	1981	Standards for air quality; CPCB
Environment (Protection) Act	1986	Umbrella act; post-Bhopal Gas Tragedy; Central Gov power to regulate
Biological Diversity Act	2002	Access and benefit sharing; NBA, SBBs, BMCs
Forest Rights Act (FRA)	2006	Rights of tribal communities over forest land/resources
National Green Tribunal Act	2010	NGT established; fast-track environmental cases

14.3 International Conventions

Convention	Year	Subject
Ramsar Convention	1971	Protection of Wetlands of International Importance
CITES	1973	International trade in endangered species (flora & fauna)
Bonn Convention (CMS)	1979	Migratory species conservation
Vienna Convention + Montreal Protocol	1985 / 1987	Ozone layer protection; phase-out of CFCs
Basel Convention	1989	Control of hazardous wastes & disposal
Convention on Biological Diversity (CBD)	1992	Biodiversity conservation; access and benefit sharing
UNFCCC	1992	Framework for combating climate change

Convention	Year	Subject
Kyoto Protocol	1997	GHG reduction targets (developed countries)
Paris Agreement	2015	Limit global warming to 1.5-2°C above pre-industrial levels; NDCs
Kunming-Montreal GBF	2022	Biodiversity targets; 30x30 protection goal

14.4 Government Schemes Related to Environment & Geography

Scheme	Key Details
Project Tiger (1973)	Conservation of Bengal Tiger; 54 reserves; NTCA oversight
Project Elephant (1992)	Elephant conservation; 33 elephant reserves
Project Crocodile (1975)	3 crocodile species protection (Mugger, Saltwater, Gharial)
Project Dolphin (2020)	Gangetic Dolphin (river) + Oceanic Dolphin conservation
CAMPA	Compensatory Afforestation Fund; Rs 40,000 cr+ collected
National Water Mission	One of 8 missions under National Action Plan on Climate Change (NAPCC 2008)
Namami Gange Programme	Integrated Ganga River rejuvenation (₹20,000 crore); 2015
AMRUT	Atal Mission for Rejuvenation and Urban Transformation; urban water infrastructure
PM Van Dhan Yojana	Forest produce for tribal livelihoods
Cheetah Reintroduction	Cheetahs from Namibia introduced at Kuno NP (MP) — 2022

EXAM FOCUS

India's National Animal = Bengal Tiger | National Bird = Indian Peacock | National Flower = Lotus
 National Aquatic Animal = Ganges River Dolphin | National Tree = Banyan | National Fruit = Mango
 Project Tiger = 1973 (India's first major conservation project); PM Indira Gandhi
 Cheetah Reintroduction = Kuno NP, MP (2022) — Namibia + South Africa cheetahs
 Gharial = Critically Endangered; found in Chambal, Girwa (Katarniaghat WS, UP), Mahanadi
 Great Indian Bustard = Critically Endangered; Rajasthan; affected by power lines

CHAPTER 15: WORLD GEOGRAPHY — QUICK REFERENCE

15.1 Continents

Continent	Area (sq km)	Largest Country	Highest Point	Lowest Point	Notes
Asia	44.6 million	Russia	Everest (8848m, Nepal/China)	Dead Sea	Largest continent; most populous
Africa	30.4 million	Algeria	Kilimanjaro (5895m, Tanzania)	Lake Assal (Djibouti)	2nd largest; Sahara (largest hot desert)
North America	24.7 million	Canada	Denali (6190m, Alaska/USA)	Death Valley (California)	Includes Central America
South America	17.8 million	Brazil	Aconcagua (6961m, Argentina)	Laguna del Carbón	Amazon River; Atacama Desert
Antarctica	14 million	—	Vinson Massif (4892m)	—	Coldest; 70% of world's fresh water (ice)
Europe	10.5 million	Russia (partly)	Mont Blanc (4808m, France/Italy)	Caspian Sea shore	Smallest continent (excl Australia)
Australia	7.7 million	Australia	Mawson Peak / Mt Kosciuszko	Lake Eyre	Smallest continent; also a country

15.2 Major Deserts of the World

Desert	Continent	Type	Key Features
Sahara	Africa	Hot	Largest hot desert; covers ~9.2 million sq km; 11 countries
Arabian Desert	Asia	Hot	Rub al Khali (Empty Quarter) — largest continuous sand desert
Gobi	Asia	Cold	Mongolia + China; rocky desert; fossil-rich
Patagonian	S America	Cold	Argentina; largest cold desert in S America
Great Australian	Australia	Hot	Sandy; largest desert in Australia
Atacama	S America (Chile)	Cold coastal	Driest non-polar desert; Chilean coast
Kalahari	S Africa	Semi-arid	Botswana + S Africa; wildlife rich

Desert	Continent	Type	Key Features
Thar	S Asia (India/Pak)	Hot	Great Indian Desert; 7th largest
Antarctic	Antarctica	Cold	Largest cold desert in world (~14 million sq km)

★ Frequently Asked in Exams

Largest hot desert = Sahara | Largest cold desert = Antarctic

Driest desert (non-polar) = Atacama (Chile)

Arabian desert contains Rub al Khali — world's largest unbroken sand sea

Gobi Desert = Mongolia/China — fossil discoveries (dinosaur eggs)

Thar = 7th largest desert in world; 'Sea of Sand' (Marusthali)

15.3 Major Mountain Ranges of the World

Mountain Range	Continent	Highest Peak	Notes
Himalayas	Asia	Everest (8848m)	Includes K2 (2nd), Kanchenjunga (3rd); youngest fold mountains
Andes	South America	Aconcagua (6961m)	Longest mountain range; Pacific coast
Rockies	North America	Elbert (4399m)	Western USA/Canada; Continental Divide
Alps	Europe	Mont Blanc (4808m)	France/Switzerland/Italy; Rhine, Rhone, Po rivers originate
Atlas	Africa	Toubkal (4167m)	NW Africa; Morocco/Algeria/Tunisia
Ural Mountains	Asia/Europe boundary	Narodnaya (1895m)	Divides Asia from Europe
Caucasus	Europe/Asia	Elbrus (5642m)	Highest peak in Europe (if Caucasus = Europe)
Appalachian	North America	Mitchell (2037m)	Eastern USA; oldest mountain range in N America
Great Dividing Range	Australia	Kosciuszko (2228m)	Australia's main mountain barrier

15.4 Ocean Currents

Current	Ocean	Type (W/C)	Effect
Gulf Stream	Atlantic	Warm	Keeps W Europe mild; world's strongest ocean current

Current	Ocean	Type (W/C)	Effect
Labrador Current	Atlantic	Cold	Meets Gulf Stream; fog off Newfoundland; good fishing
Canary Current	Atlantic	Cold	SW of Europe/NW Africa; N-bound cold current
N Atlantic Drift	Atlantic	Warm	Extension of Gulf Stream; keeps UK warm
Brazil Current	Atlantic	Warm	South American coast; warm
Benguela Current	Atlantic	Cold	W coast of S Africa; upwelling; rich fish
Kuroshio Current	Pacific	Warm	Japan; NE Pacific; warm
North Pacific Current	Pacific	Warm	Extension of Kuroshio
California Current	Pacific	Cold	W USA coast; cold
Peru/Humboldt Current	Pacific	Cold	W South America; anchoveta fishing; El Nino affects it
Indian Ocean Currents	Indian Ocean	Seasonal	Reverse with monsoon; N. Equatorial, Somali Current (changes direction with monsoon)

🔥 EXAM FOCUS

Warm currents flow from equator to poles; Cold currents from poles to equator
 Warm current coasts = moist/mild climate | Cold current coasts = fog/desert (Atacama, Namib)
 Gulf Stream = warms Western Europe | Labrador = creates Grand Banks fishing
 Peru/Humboldt Current → disrupted by El Nino → drought in India
 Somali Current (Indian Ocean) reverses direction with monsoon season

15.5 Major Rivers of the World

River	Continent	Length	Key Facts
Nile	Africa	~6650 km	Longest river in world; flows N into Mediterranean
Amazon	S America	~6400 km	Largest by discharge; tropical rainforest basin
Yangtze	Asia	~6300 km	Longest river in Asia; 3 Gorges Dam
Mississippi-Missouri	N America	~6275 km	Longest in N America
Yenisei	Asia	~5539 km	Siberia; flows into Arctic Ocean
Yellow River (Huang He)	Asia	~5464 km	Cradle of Chinese civilization; 'China's Sorrow'
Ob-Irtysh	Asia	~5410 km	Siberia; drains into Arctic
Congo	Africa	~4700 km	2nd largest by discharge; deepest river
Amur	Asia	~4444 km	Russia-China border
Mekong	Asia	~4350 km	SE Asia; passes 6 countries; Lancang in China

River	Continent	Length	Key Facts
Volga	Europe	~3690 km	Longest river in Europe; Russia; Caspian Sea
Rhine	Europe	~1230 km	W Europe; major navigable river; Alps to North Sea
Danube	Europe	~2860 km	2nd longest in Europe; flows E into Black Sea
Zambezi	Africa	~2574 km	Victoria Falls; Mozambique channel
Irrawaddy	Asia	~2170 km	Myanmar; drains into Andaman Sea

★ Frequently Asked in Exams

Nile (Africa) = Longest river in world (~6650 km)

Amazon (S America) = Largest river by volume/discharge

Yangtze (China) = Longest river in Asia; Three Gorges Dam on it

Volga (Russia) = Longest river in Europe

Congo (Africa) = Deepest river in the world

Victoria Falls = on Zambezi River (Zimbabwe/Zambia border) — largest waterfall by volume

CHAPTER 16: MASTER ONE-LINERS FOR LAST-MINUTE REVISION

India — Basic Geography

- ⚡ India's Standard Meridian = 82.5°E → passes through Mirzapur (UP)
- ⚡ India's latitudinal extent: $8^{\circ}4'\text{N}$ to $37^{\circ}6'\text{N}$ (mainland); Longitudinal: $68^{\circ}7'\text{E}$ to $97^{\circ}25'\text{E}$
- ⚡ N-S distance = 3,214 km; E-W distance = 2,933 km
- ⚡ India is in Northern + Eastern hemisphere
- ⚡ Tropic of Cancer (23.5°N) passes through 8 states: Guj, Raj, MP, CG, Jharkhand, WB, Tripura, Mizoram
- ⚡ IST = GMT + 5:30 hours (82.5°E Standard Meridian)

Mountains & Passes

- ⚡ Highest peak in India = K2 (if J&K claimed) OR Kanchenjunga (8586m) in Indian territory
- ⚡ Highest peak in Peninsular India = Anaimudi (2695m) — Western Ghats, Kerala
- ⚡ Highest peak in Aravalli = Guru Shikhar (1722m) — Rajasthan
- ⚡ Aravalli = Oldest mountain range in India
- ⚡ Western Ghats + Eastern Ghats meet at Nilgiris (TN/Kerala)
- ⚡ Satpura = between Narmada (N) and Tapti (S)
- ⚡ Nanda Devi (7816m) = Highest peak in India in fully Indian territory (Uttarakhand)

Rivers

- ⚡ Ganga = National River; Gangetic Dolphin = National Aquatic Animal
- ⚡ Yamuna = longest tributary of Ganga
- ⚡ Kosi = Sorrow of Bihar (frequent flooding)
- ⚡ Godavari = Dakshin Ganga (largest peninsular river)
- ⚡ Narmada + Tapti = only major west-flowing rift valley rivers
- ⚡ Mahi = crosses Tropic of Cancer TWICE
- ⚡ Jog Falls (Sharavati, Karnataka) = highest plunge waterfall in India
- ⚡ Brahmaputra = world's largest sediment-carrying river; Majuli = largest river island
- ⚡ Indus Waters Treaty (1960) = India gets E rivers; Pakistan gets W rivers
- ⚡ Chambal = originates from Vindhya Range (Janapav Hill, MP)
- ⚡ Sabarmati = Gandhi Ashram at Ahmedabad; flows into Gulf of Khambhat
- ⚡ Luni = only river of Rajasthan reaching sea (becomes saline after Balotra)
- ⚡ Sharavati = Jog Falls | Narmada = Dhuandhar Falls (Jabalpur/Bhedaghat)

Soils & Forests

- ⚡ Black soil = best for cotton; self-ploughing; from Deccan basalt

- ⚡ Laterite soil = high rainfall areas; leaching removes Ca and Si; Fe and Al remain
- ⚡ Alluvial soil = most widespread in India; most fertile; khadar > bangar
- ⚡ Sundarbans = largest mangrove forest in the world
- ⚡ Western Ghats = UNESCO World Heritage Biodiversity Hotspot
- ⚡ Mawsynram (Meghalaya) = wettest place in India and world
- ⚡ Largest state by forest area = Madhya Pradesh

Lakes & Dams

- ⚡ Wular Lake (J&K) = largest freshwater lake in India
- ⚡ Chilika Lake (Odisha) = largest brackish water lagoon in India; 1st Ramsar site
- ⚡ Sambhar Lake (Rajasthan) = largest inland saltwater lake
- ⚡ Loktak Lake (Manipur) + Keibul Lamjao NP = world's only floating national park
- ⚡ Lonar Lake (Maharashtra) = meteorite crater lake; saline + alkaline
- ⚡ Pangong Tso (Ladakh) = 60% in China; ~4350m altitude
- ⚡ Tehri Dam = highest dam in India (~261m) | Hirakud = longest dam
- ⚡ Bhakra Nangal = tallest gravity dam in Asia; reservoir = Gobind Sagar
- ⚡ Indira Sagar Dam (Narmada, MP) = largest reservoir in India by capacity

Wildlife & Environment

- ⚡ Jim Corbett = India's first national park (1936); 1st Project Tiger park (1973)
- ⚡ Kaziranga = largest population of one-horned rhinos; UNESCO WH
- ⚡ Gir Forest = only habitat of Asiatic Lion in wild
- ⚡ Keibul Lamjao NP = world's only floating NP (Manipur)
- ⚡ Barren Island = India's only active volcano (Andaman)
- ⚡ Narcondam Island = dormant volcano (Andaman)
- ⚡ 10° channel = divides Andaman from Nicobar
- ⚡ Lakshadweep = coral islands; 36 islands; smallest UT; 9° channel separates from Minicoy
- ⚡ India has 54 Tiger Reserves + 75+ Ramsar Sites
- ⚡ Project Tiger = 1973 | Project Elephant = 1992 | Project Crocodile = 1975
- ⚡ Cheetah reintroduction = Kuno NP, Madhya Pradesh (2022)

Neighbors & Borders

- ⚡ Bangladesh = longest land border with India
- ⚡ China shares border with 5 states: J&K, HP, Uttarakhand, Sikkim, Arunachal
- ⚡ Treaty of Sugauli (1816) = Anglo-Nepalese War; Kali River = W Nepal boundary
- ⚡ 100th Constitutional Amendment (2015) = India-Bangladesh land boundary agreement
- ⚡ Manas NP (Assam) extends into Bhutan as Royal Manas NP
- ⚡ Kalapani = Uttarakhand (Pithoragarh district); disputed with Nepal

- ⚡ Mount Saramati = highest peak of Nagaland (Naga Hills)
- ⚡ Palk Strait separates India from Sri Lanka; Adam's Bridge (Ram Setu) = limestone shoals

END OF DOCUMENT — BEST OF LUCK FOR YOUR EXAMINATION!